

Notes: Schenone (RFS, 2010)

Lending Relationships and Information Rents: Do Banks Exploit Their Information Advantages?

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1 Big picture

- **Question.** Do relationship banks exploit the firm-specific information they acquire through lending relationships by charging borrowers higher loan rates than would prevail if all banks were symmetrically informed?
- **Setting.** The paper studies syndicated bank loans to U.S. firms around a major information event: the firm's initial public offering (IPO). The core comparison is between the pricing of loans **before** and **after** the IPO.
- **Core challenge.** A lending relationship can change borrowing costs for two opposite reasons. On the one hand, repeated lending can lower the lender's monitoring and screening cost. On the other hand, it can give the incumbent lender an information monopoly that raises the borrower's switching cost and allows rent extraction.
- **Main methodological contribution.** The paper combines a new measure of **relationship intensity** with a panel of firm-loan observations around the IPO. The IPO is treated as an information-releasing event that weakens the incumbent bank's information monopoly. This lets the paper compare how the relationship-pricing pattern changes when outside lenders become better informed.
- **Why it matters.** If banks extract informational rents from locked-in borrowers, then the cost of bank finance depends not just on risk and competition, but also on private information generated inside lending relationships. This matters for corporate financing choices and for the firm's decision to go public.
- **Main empirical question.** How does the relationship between loan spreads and relationship intensity differ before and after a firm's IPO?
- **Main empirical results.**
 - Prior to the IPO, the relation between borrowing rates and relationship intensity is **U-shaped**.
 - After the IPO, borrowing rates are **monotonically decreasing** in relationship intensity.
 - In raw summary statistics, the mean spread on drawn funds falls from **248.22** basis points above LIBOR before the IPO to **227.01** after the IPO, a drop of **21.21** basis points.
 - In fixed-effects specifications that compare loans before and after the IPO within firms, the post-IPO spread falls by roughly **50–71 basis points**, depending on the sample restriction.
 - The results are robust to controls for leverage, debt ratings, loan type, loan purpose, firm age, and loan-year fixed effects.
 - Alternative explanations based on **risk sharing** in larger syndicates and on **concurrent lending and underwriting** do not overturn the main interpretation.
- **Economic takeaway.** Before public disclosure, relationship banks appear able to exploit an information-based monopoly over borrowers. Once the IPO disseminates information to the market, the incumbent lender loses that monopoly and rates become more competitive.

2 Data and measurement (key objects)

2.1 Sample construction and data sources

- **IPO sample.** IPO firms are identified using the *Securities Data Company* (SDC) database. The paper keeps U.S. IPOs between **1998 and 2003**, excluding ADRs, REITs, financial institutions, private placements, rights issues, unit issues, best-efforts offerings, nonfirm commitment offers, and auctions.
- **Loan data.** These IPOs are hand-matched to loans in *Dealscan*. Dealscan reports loan characteristics for loans above **\$100,000**, including spreads, fees, maturity, syndicate structure, lender identities, loan type, loan purpose, and credit ratings.
- **Firm accounting data.** To control for risk around the IPO, the paper uses *COMPUSTAT* and the firm's **IPO prospectus** to construct leverage, debt, asset, and equity measures in the year the loan was taken.
- **Baseline sample size.**
 - Dealscan reports at least one loan for **411** IPO firms.
 - For these firms, there are **981** loans with all-in-spread-drawn available.
 - Of these, **511** loans are pre-IPO and **470** are post-IPO.
 - **250** firms have at least one loan before and one loan after the IPO.
- **Selection issue and interpretation.** The paper argues that the sample is, if anything, biased *against* finding information-rent extraction because IPO firms are relatively mature and transparent compared with the typical private firm, and Dealscan mainly covers relatively large loans.

2.2 Unit of observation and key loan objects

- **Unit of observation.** The empirical observation is a **firm-loan** observation, interpreted as one step in a firm's evolving lending relationship.
- **Main pricing variable.** The core dependent variable is **all-in-spread-drawn**, the amount the borrower pays in basis points above LIBOR on drawn funds. It incorporates the loan spread plus annual or facility fees allocated to drawn amounts.
- **Other loan-price objects.**
 - all-in-spread-undrawn,
 - up-front fees,
 - commitment fees.
- **Other loan characteristics.**
 - loan amount,
 - maturity,
 - whether the loan is secured,

- number of lending banks,
- percentage of the loan contributed by the lead lender,
- loan type (revolver, term loan, 364-day facility, etc.),
- loan purpose (debt repayment, takeover, acquisition, LBO/MBO, etc.),
- S&P senior debt rating at origination.

2.3 Relationship intensity and lock-in

The paper's central innovation is a measure of how dependent a firm is on its current lead lender:

$$\text{Intensity}_{ij} = \frac{\text{Prior loans by the current lead lender up to loan } j}{\text{Total number of loans taken by firm } i \text{ up to loan } j}.$$

- For example, if a firm has taken 10 loans to date and 7 of them involve the current lead lender, then relationship intensity with that lender is **0.7**.
- This measure differs from the standard **loan maturity** proxy. Loan maturity captures how long a loan lasts, but not how concentrated the borrower's financing is with one lender.
- The paper also defines:
 - **Switched lender**: an indicator equal to 1 if the current lead lender did not lead any prior loan to the firm;
 - **First loan**: an indicator for the first observed loan;
 - an alternative **amount-based intensity** measure using the cumulative dollar amount borrowed from the current lead lender as a fraction of total cumulative borrowing.
- The construction of relationship intensity explicitly accounts for **bank mergers** and for loans granted by **subsidiaries** of a bank holding company, treating information as likely shared within the merged or affiliated institution.

2.4 Key descriptive facts

- The mean all-in-spread-drawn in the full sample is **238.06** basis points above LIBOR.
- Before versus after the IPO:
 - pre-IPO spread on drawn funds: **248.22** bps,
 - post-IPO spread on drawn funds: **227.01** bps,
 - difference: **-21.21** bps.
- Syndicate structure changes around the IPO:
 - number of lending banks rises from **4.96** to **6.10**,
 - the lead lender's share falls from **67.63%** to **55.26%**.
- Average loan amount rises from **\$190.52 million** pre-IPO to **\$231.16 million** post-IPO.
- Average maturity falls from **46.57** months pre-IPO to **43.11** months post-IPO.

2.5 Information structure and timing

- **Before the IPO.** Firms are private, disclosure is limited, and the relationship bank's accumulated private information can create an information monopoly.
- **During and after the IPO.** The IPO prospectus, SEC scrutiny, ongoing public disclosure requirements, and the stock price release information to outsiders.
- **Implication.** The paper interprets the IPO as a large information shock that levels the playing field among current and prospective lenders.
- **Important caveat.** Some firms issue public or private debt before the IPO, which could already reveal information. The paper studies this possibility explicitly and finds that such cases are relatively limited and do not drive the core results.

3 Models and empirical specifications (all equations + interpretation)

3.1 Theoretical setup: lending costs and information asymmetry

- **Relationship-lender cost.** As the same bank lends repeatedly to a firm, its cost of lending falls because monitoring, due diligence, and interpretation of borrower-specific information become more efficient.
- **Outside-lender cost.** The cost faced by outside banks depends on the extent of **asymmetric information** relative to the relationship lender.
- **Key idea.** The observed borrowing rate reflects the interaction of two forces:
 - **information spillovers**, which reduce the incumbent lender's monopoly by revealing information to outsiders, and
 - **information-processing capacity**, which strengthens the incumbent lender's advantage as it becomes better at gathering and using borrower-specific information.

Interpretation. If spillovers dominate, rates should fall with relationship intensity. If information-processing capacity dominates, rates should stop falling and may even rise. This is the origin of the paper's pre-IPO **U-shaped** prediction.

3.2 Pre-IPO regression

For pre-IPO loans, the main specification is:

$$\text{Spread}_{ij} = \beta_0 + \beta_I \text{Intensity}_{ij} + \beta_{I^2} \text{Intensity}_{ij}^2 + \beta_X X_{ij} + \beta_F F_{ij} + \beta_S \text{Switched}_{ij} + \beta_1 \text{FirstLoan}_{ij} + \delta_{t(j)} + \alpha_i + u_{ij}.$$

- X_{ij} : loan characteristics such as maturity, loan type, loan purpose, and syndicate size.
- F_{ij} : firm characteristics such as debt rating and age.
- $\delta_{t(j)}$: loan-year fixed effects.

- α_i : firm fixed effects.

Interpretation. The coefficient on Intensity_{ij} captures how the price of borrowing changes as the firm becomes more dependent on the current lender, using only within-firm variation. The quadratic term allows the effect to turn from negative to positive as lock-in deepens.

In the baseline pre-IPO regression, the paper estimates:

$$\hat{\beta}_I = -474.55, \quad \hat{\beta}_{I^2} = 402.00,$$

with a turning point at roughly **0.59**. That is, for relationship intensity below about 0.59, spreads fall as intensity rises; above 0.59, spreads rise. The paper interprets this as banks beginning to exploit information rents once they have provided about **60%** of the firm's loans.

3.3 Post-IPO regression

For post-IPO loans, the same idea is estimated on the subsample of post-IPO observations. Empirically, the quadratic term is not important, and the relevant relationship is approximately linear:

$$\text{Spread}_{ij} = \beta_0 + \beta_I \text{Intensity}_{ij} + \beta_X X_{ij} + \beta_F F_{ij} + \delta_{t(j)} + \alpha_i + u_{ij}.$$

Interpretation. After the IPO, the lender's private information advantage is weakened by public disclosure. Relationship intensity now mainly captures the lower operating cost of repeat lending, not monopoly power. Accordingly, the spread declines monotonically in intensity.

A representative specification with leverage controls yields a post-IPO intensity coefficient of about **-40.11**, consistent with a downward-sloping relation between spreads and relationship intensity after the IPO.

3.4 Full before-versus-after IPO specification

To compare the two regimes directly, the paper estimates:

$$\text{Spread}_{ij} = \beta_A(\text{Intensity}_{ij} \times \text{AfterIPO}_{ij}) + \beta_{A2}(\text{Intensity}_{ij} \times \text{AfterIPO}_{ij})^2 + \beta_B(\text{Intensity}_{ij} \times \text{BeforeIPO}_{ij}) + \beta_{B2}(\text{Intensity}_{ij} \times \text{BeforeIPO}_{ij})^2 + \beta_0 + \beta_X X_{ij} + \beta_F F_{ij} + \delta_{t(j)} + \alpha_i + u_{ij}.$$

Interpretation. This specification asks two separate questions at once:

- before the IPO, is the spread-intensity pattern U-shaped?
- after the IPO, does that pattern become monotone decreasing?

The coefficient on AfterIPO captures the average shift in spreads after the firm goes public, while the interaction terms describe how the slope and curvature of the relationship change.

3.5 Alternative measure of relationship intensity

The paper also uses a dollar-based dependence measure:

$$\text{Intensity}_{ij}^{\text{Amount}} = \frac{\text{Total amount borrowed from the current lead lender to date}}{\text{Total amount borrowed by the firm to date}}.$$

Interpretation. This checks whether the results depend on counting loans rather than loan volumes. The main pattern remains: pre-IPO rates rise with dependence, while post-IPO rates fall.

3.6 Risk-sharing specification

To test whether larger syndicates after the IPO explain the spread decline, the paper estimates regressions of the form:

$$\text{Spread}_{ij} = \beta_0 + \beta_P \text{AfterIPO}_{ij} + \beta_R \text{RiskSharing}_{ij} + \beta_{RP} (\text{RiskSharing}_{ij} \times \text{AfterIPO}_{ij}) + \alpha_i + \delta_{t(j)} + u_{ij},$$

where risk sharing is measured by the number of lenders, the lead lender's percentage share, or the Herfindahl concentration of the syndicate.

Interpretation. If lower post-IPO spreads are mainly due to better diversification of loan risk, these variables should explain the rate drop and should weaken the role of relationship intensity.

3.7 Concurrent lending-underwriting specification

To test whether banks cut loan rates in exchange for IPO underwriting business, the paper estimates:

$$\text{Spread}_{ij} = \beta_0 + \beta_P \text{AfterIPO}_{ij} + \beta_U \text{LeadLenderIPOManager}_{ij} + \beta_{UP} (\text{LeadLenderIPOManager}_{ij} \times \text{AfterIPO}_{ij}) + \alpha_i + \delta_{t(j)} + u_{ij},$$

and analogous regressions replacing IPO manager with IPO book runner.

Interpretation. If post-IPO lending rates are lower because the bank is being rewarded with underwriting mandates, then the underwriting-role variables should explain the pattern that the paper attributes to information rents.

4 Identification and instruments (what makes the estimates credible)

4.1 What standard approaches miss

- Traditional work often compares borrowing costs across firms with longer and shorter loan maturities in a cross section.
- Such comparisons mix together firm heterogeneity, lender learning, switching costs, and the firm's endogenous choice of relationship structure.
- They also do not cleanly separate the lender's changing cost of lending from the lender's changing information monopoly.

4.2 What identifies the paper's approach

- **The IPO as an information shock.** The key identifying idea is that the IPO credibly releases information to the public market and sharply reduces the incumbent lender's private informational advantage.
- **Within-firm variation.** Firm fixed effects absorb time-invariant firm traits such as managerial quality, governance, negotiation ability, and persistent differences in transparency.
- **Loan-year fixed effects.** These absorb common shifts in credit conditions across loans originated in the same year.
- **Relationship-intensity variation.** The coefficient estimates are identified from how a given firm's spreads change as it becomes more or less dependent on a given lender, and from how that pattern differs before and after the IPO.

4.3 Why the IPO is useful here

- Before going public, firms do not face the same systematic disclosure obligations as public firms.
- During the IPO, the prospectus and SEC scrutiny release substantial information.
- After the IPO, regular disclosure plus the stock price keep outside lenders better informed.

Interpretation. The IPO is not used as a shock to cash flows or technology. It is used as a shock to the **distribution of information across lenders**.

4.4 Why the estimates are not just picking up risk changes

- The paper uses S&P senior debt ratings at loan origination.
- It adds firm leverage measured in the year the loan was taken, using COMPUSTAT and IPO prospectuses.
- It checks directly whether ratings distributions change before and after the IPO and finds little evidence that they do.

Interpretation. The post-IPO spread decline is not simply a mechanical effect of firms becoming safer after going public.

4.5 Why alternative explanations do not overturn the result

- **Risk sharing.** Post-IPO syndicates do become larger and less concentrated, but once relationship-intensity controls are added, the risk-sharing measures lose much of their explanatory power while the relationship variables remain strong.
- **Concurrent underwriting.** Although banks that underwrite the IPO are correlated with lower loan spreads in raw comparisons, the underwriting-role variables do not explain away the before-versus-after IPO relationship pattern.

Table 1
Summary statistics for key loan characteristics

	Overall sample (1)	Before IPO (2)	After IPO (3)	Difference (4)
Interest rates				
All-in-spread drawn	238.06 (111.38)	248.22 (106.81)	227.01 (115.25)	-21.21*** (7.11)
Interest rate spread on drawn funds, in basis points above LIBOR	981	511	470	981
All-in-spread un-drawn (basis points total)	41.74 (28.62)	42.80 (31.24)	40.81 (26.13)	-2.00 (2.30)
	640	298	342	640
Fees				
Up-front fees (basis points)	60.72 (59.08)	57.11 (52.96)	67.63 (69.07)	10.52 (7.81)
	297	195	102	297
Commitment fees (basis points)	46.17 (28.16)	46.80 (31.39)	45.61 (24.80)	-1.19 (2.49)
	526	248	278	526
More loan characteristics				
Secured (1 if secured, 0 o/w)	0.86 (0.35)	0.88 (0.32)	0.84 (0.37)	-0.04* (0.03)
	688	372	316	688
Loan amount (millions of U.S. dollars)	209.92 (386.74)	190.52 (412.48)	231.16 (355.67)	40.64* (24.39)
	993	519	474	993
Maximum loan maturity (months)	44.91 (28.15)	46.57 (30.80)	43.11 (24.85)	-3.47** (1.77)
	989	515	474	989
Number lending banks	5.51 (6.78)	4.96 (7.37)	6.10 (6.04)	1.14*** (0.43)
	981	508	473	981
Pct lead lender lent	61.63 (37.03)	67.63 (36.50)	55.26 (36.55)	-12.38*** (2.61)
	786	405	381	786

The entries in each cell correspond to the mean, standard deviation in parentheses, and number of observations for each loan characteristic. The interest rates for the loans (all-in-spread drawn, all-in-spread undrawn, and LIBOR), and all fees (commitment and up-front fees) are specified in basis points. When a loan has several facilities, these interest rates and fees are the weighted average of the interest rates of each facility where the weights are the amount of the loan in that facility relative to the total loan amount. Loan maturity is measured in months, and for the loans where there are multiple facilities, the length of the loan corresponds to the maximum between the lengths of the facilities. I use ***, **, and * to denote significance at the 1%, 5%, and 10% levels, respectively.

- **Tech and internet bubble firms.** Including or excluding tech and internet IPOs does not overturn the core findings.
- **Earlier debt issues.** Very few firms issue public debt before the IPO, and only a limited set place private debt before the IPO; excluding these firms leaves the results essentially unchanged.

4.6 Relation to the literature

- Relative to the relationship-lending literature, the paper replaces maturity with a more direct measure of **dependence on the current lender**.
- Relative to cross-sectional studies, it contributes a **panel design with firm fixed effects**.
- Relative to the IPO literature, it highlights a financing-side benefit of going public: public disclosure can weaken the information monopoly of relationship banks.

5 Tables and figures

5.1 Table 1: Summary statistics for loan characteristics

- **Read:** This table establishes the main before-versus-after IPO facts in raw data.
- The spread on drawn funds falls by **21.21 bps** after the IPO.
- The number of lending banks rises by **1.14** on average.
- The lead lender's share of the loan falls by **12.38 percentage points**.

- **Takeaway.** Public firms appear to borrow in more competitive syndicates and at lower rates.

5.2 Table 2: Pre-IPO regressions

Table 2
Cost of loans, as a function on relationship intensity, before the firm's IPO

Panel A: Baseline regressions and loan characteristics

	Baseline (1)	Baseline (2)	Maturity (3)	Maturity × Intensity (4)
Relationship variables				
Number of prior loans by lead				
Total number of loans to date	-474.55*** (179.87)	-385.91** (178.56)	-417.75** (174.84)	-416.63** (175.19)
(Number of prior loans by lead) ²	402.00** (172.38)	303.94* (171.79)	351.42** (168.61)	367.00** (171.42)
Total number of loans to date	-98.26** (44.24)	-80.37* (43.86)	-85.64** (42.97)	-87.58** (43.20)
Switched lenders				
Loan characteristics				
First loan	-121.83*** (45.96)	-98.15** (45.65)	-102.11** (44.65)	-104.56* (44.97)
Number of lending banks				
Loan maturity				
Loan maturity × Intensity				
Number of prior loans by lead				
Total number of loans to date				
Firm fixed effects	Yes	Yes	Yes	Yes
Loan year fixed effects	No	Yes	Yes	Yes
R ² within	0.05	0.13	0.18	0.18
F-statistic	2.57	2.67	3.42	3.16
Number of firms	98	98	97	97
Number of loans	295	295	293	293
Average number of loans	3.01	3.01	3.02	3.02
Max. number of loans per firm	2	2	2	2
Min. number of loans per firm	9	9	9	9

(continued overleaf)

Table 2
(Continued)

Panel B: More loan and firm characteristics

	Loan type (1)	Loan type (2)	Loan purpose (3)	Rating (4)	Firm age (5)
Relationship variables					
Number of prior loans by lead					
Total number of loans to date	-348.62** (172.26)	-331.27* (173.65)	-370.34** (169.78)	-377.31** (170.17)	-394.21** (175.28)
(Number of prior loans by lead) ²	276.70* (166.49)	268.15* (166.95)	303.29* (164.25)	314.41* (164.73)	327.18* (169.35)
Total number of loans to date	-71.68* (42.15)	-70.60* (42.20)	-82.04** (41.09)	-83.13** (41.16)	-87.55** (42.58)
Switched lenders					
Loan characteristics					
First loan	-84.60* (44.06)	-86.32* (44.14)	-95.97** (42.95)	-97.53** (43.03)	-103.02** (45.09)
Number of lending banks					
Loan maturity					
Loan maturity × Intensity					
Number of prior loans by lead					
Total number of loans to date					
Loan type					
Term loan	9.36 (15.77)	11.91 (16.07)	18.36 (15.50)	17.64 (15.54)	17.13 (15.63)
Revolver	-30.65*** (11.29)	-21.61 (15.67)	-26.07** (11.10)	-25.70** (11.12)	-26.00** (11.17)
Revolver × Intensity					
Number of prior loans by lead					
Total number of loans to date					
Loan purpose					
Debt repayment					
Takeover					
Acquisition					
LBOMBO					
Firm characteristics					
S&P Senior rating:					
Vulnerable					
Firm years at relationship					
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Loan year fixed effects	Yes	Yes	Yes	Yes	Yes
R ² within	0.23	0.23	0.30	0.30	0.30
F-statistic	3.90	3.68	4.19	3.99	3.78
Number of firms	97	97	97	97	97
Number of loans	293	293	293	293	293
Average number of loans	3.02	3.02	3.02	3.02	3.02
Max. number of loans per firm	9	9	9	9	9
Min. number of loans per firm	2	2	2	2	2

(continued overleaf)

Table 2
(Continued)

Panel C: More relationship variables: Link between subsequent lenders and between the lead lender and the IPO underwriter

	Firm kept prior Lender (1)	Lender (2)
Relationship variables		
Number of prior loans by lead		
Total number of loans to date	-358.92** (168.59)	-340.35** (170.80)
(Number of prior loans by lead) ²	298.56* (163.19)	281.41* (164.43)
Total number of loans to date	-83.19* (41.28)	-83.31* (40.96)
Switched lenders		
More relationship variables		
Kept immediate prior lead lender		
Kept any of the immediate prior lender		
Loan characteristics		
First loan		
Number of lending banks		
Loan maturity		
Loan type		
Revolver		
Loan purpose		
Debt repayment		
Acquisition		
LBOMBO		
Firm characteristics		
S&P Senior rating:		
Vulnerable		
Firm fixed effects	Yes	Yes
Loan year fixed effects	Yes	Yes
R ² within	0.30	0.30
F-statistic	4.02	4.00
Number of firms	97	97
Number of loans	293	293
Average number of loans	3.02	3.02
Max. number of loans per firm	9	9
Min. number of loans per firm	2	2

Only pre-IPO loans, and only firms with a minimum of two such loans, are included in the final regression sample. The reported results correspond to fixed effects estimates:

$$\text{Spread}_{i,t} = \beta_0 + \beta_{\text{Intensity}} \text{Intensity}_{i,t} + \beta_{\text{SwitchedLenders}} \text{SwitchedLenders}_{i,t} + \beta_{\text{LoanType}} \text{LoanType}_{i,t} + \beta_{\text{LoanPurpose}} \text{LoanPurpose}_{i,t} + \beta_{\text{LoanMaturity}} \text{LoanMaturity}_{i,t} + \beta_{\text{LoanYear}} \text{LoanYear}_{i,t} + \epsilon_{i,t} + \mu_{i,t}$$

where i indexes for firm and t for the loan number. Intensity _{i,t} = $\frac{\text{Number of prior loans by lead}}{\text{Total number of loans to date}}$. Spread _{i,t} is the all-in spread down for firm i 's loan t . When a loan has several facilities, the spread is the weighted average of the spread of each facility, where the weights are the amount of the loan in that facility relative to the total loan amount, and the time to maturity is the maximum between the maturities of each facility. SwitchedLenders _{i,t} is a binary variable equal to 1 when the firm drops all previous lenders and borrows from a new lender. FirstLoan _{i,t} is a binary variable equal to 1 when firm i 's loan t is the firm's first loan. For these last two variables, Intensity _{i,t} = 0 as the number of prior loans by lead lender is 0. LoanYearFE _{i,t} is a vector of loan-year dummies. LoanCharacteristics _{i,t} is a vector of firm i 's loan t 's characteristics, such as time to maturity, and loan type and purpose. FirmCharacteristics _{i,t} is a vector of firm i 's characteristics at the time loan t was taken, such as firm i 's debt rating when taking loan t . Debt ratings are S&P senior debt ratings at the onset of the loan, as reported in DataComp. Panel A presents the baseline regression. Panels B and C show further robustness checks. 1 use ***, **, and * to denote significance at the 1%, 5%, and 10% levels, respectively.

- **Read:** This is the core evidence for information-rent extraction before the IPO.
- The estimated relation between spreads and relationship intensity is **U-shaped**.
- In the baseline specification, the turning point is around **0.59**: once the current lender has provided about 60% of the firm's loans, additional dependence raises spreads.
- Switching lenders lowers rates substantially, by roughly **80–98 bps** in the baseline pre-IPO specifications.
- **Takeaway.** Early in a relationship, repeated lending helps the borrower; later, deeper dependence allows the lender to extract informational rents.

5.3 Table 3: Post-IPO regressions

- **Read:** This table asks whether the same lock-in pattern survives once the firm becomes public.
- The quadratic term disappears in importance and the spread becomes **monotonically decreasing** in relationship intensity.
- In a specification with leverage controls, the post-IPO intensity coefficient is around **-40.11**.
- Switching lenders no longer has the same strong explanatory role as in the pre-IPO sample.
- **Takeaway.** After disclosure becomes public, relationship intensity mainly reflects lower lending costs, not monopoly power.

Table 3
Cost of loans, as a function of relationship intensity, after the firm's IPO

Panel A: Baseline regressions and loan characteristics

	Baseline (1)	Baseline (2)	Baseline (3)	Switched lenders (4)
Relationship variables				
Number of prior loans by lead	-111.66*	-62.94***	-53.58**	-92.13***
Total number of loans to date	(63.25)	(22.80)	(22.64)	(34.41)
(Number of prior loans by lead Total number of loans to date) ¹	70.29 (85.12)			
Switched lenders				-29.09 (19.59)
Loan characteristics				
First loan	-91.31*** (22.78)	-89.04*** (22.59)	-59.34** (25.13)	-82.64*** (29.57)
Firm fixed effects	Yes	Yes	Yes	Yes
Loan year fixed effects	No	No	Yes	Yes
R ² within	0.07	0.07	0.12	0.13
F-statistic	6.14	8.88	4.08	3.89
Number of firms	116	116	116	116
Number of loans	367	367	367	367
Average number of loans	3.16	3.16	3.16	3.16
Max. number of loans per firm	13	13	13	13
Min. number of loans per firm	2	2	2	2

(continued overleaf)

Table 3
(Continued)

Panel B: More loan and firm characteristics

	Maturity (1)	Loan purpose (2)	Loan type (3)	Rating (4)	Firm age (5)	Leverage (6)
Relationship variables						
Number of prior loans by lead	-44.66*	-43.65*	-50.09**	-49.82*	-49.48**	-40.11*
Total number of loans to date	(22.89)	(22.68)	(22.50)	(22.49)	(22.48)	(24.17)
Loan characteristics						
First loan	-57.60**	-47.26*	-48.34*	-48.28*	-51.83**	-41.20*
Number of lending banks	(24.99)	(25.07)	(24.82)	(24.81)	(24.96)	(24.62)
Loan maturity	-2.43**	-2.61**	-2.58**	-2.67**	-2.55**	-1.51
Loan purpose	(1.06)	(1.06)	(1.05)	(1.06)	(1.05)	(1.20)
Loan maturity	0.09	0.09	0.09	0.09	0.09	0.09
Loan purpose	(0.25)	(0.25)	(0.25)	(0.25)	(0.25)	(0.25)
Debt repayment	6.33	9.44	9.86	11.50	16.34	16.34
LBOMBO	(14.21)	(14.06)	(14.06)	(14.13)	(14.26)	(14.26)
Takeover	77.10	65.34	66.04	67.63	1.81	1.81
364-day facility	(51.73)	(51.23)	(51.21)	(51.22)	(76.69)	(76.69)
Firm characteristics						
S&P Senior rating: Vulnerable	71.03***	68.88**	69.17**	70.71***	83.17***	83.17***
Firm years at relationship	(27.07)	(26.75)	(26.74)	(26.77)	(30.15)	(30.15)
Loan type						
Revolver	-31.06**	-30.39**	-33.50***	-47.88***	-47.88***	-47.88***
Term loan	(11.53)	(11.54)	(11.60)	(12.81)	(12.81)	(12.81)
364-day facility	-37.61**	-37.36**	-38.67**	-23.97	-23.97	-23.97
Firm characteristics						
S&P Senior rating: Vulnerable	(18.45)	(18.44)	(18.45)	(19.30)	(19.30)	(19.30)
Firm years at relationship	-14.47	-14.19	-14.89	-9.80	-9.80	-9.80
Leverage ratio	(19.02)	(19.02)	(19.01)	(21.39)	(21.39)	(21.39)
Log (Sales)						
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Loan year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R ² within	0.14	0.16	0.20	0.20	0.26	0.26
F-statistic	3.80	3.92	3.85	3.68	3.70	3.89
Number of firms	116	116	116	116	116	96
Number of loans	367	367	367	367	367	302
Average number of loans	3.16	3.16	3.16	3.16	3.16	3.08
Max. number of loans per firm	2	2	2	2	2	2
Min. number of loans per firm	13	13	13	13	13	9

(continued overleaf)

Table 4
Changes in loan risk following the borrower's IPO

Panel A: S&P Senior debt rating at the onset of the bank loan

	(1) Not Rated	(2) Rated	(3) Above BBB	(4) Vulnerable	(5) Below vulnerable
Number of loans	Before IPO 309	Before IPO 268	Before IPO 38	Before IPO 54	Before IPO 167
	After IPO 208	After IPO 17	After IPO 58	After IPO 87	After IPO 143

Panel B: S&P senior debt rating at the onset of the bank loan

	(1) As	(2) BBBs	(3) BBs	(4) B1s	(5) CCCs	(6) CC and below
Loans	Before IPO 12	Before IPO 9	Before IPO 17	Before IPO 26	Before IPO 58	Before IPO 10
Mean	0.023	0.019	0.033	0.055	0.112	0.116
Std dev	(0.150)	(0.137)	(0.178)	(0.228)	(0.315)	(0.321)

Panel C: Secured and unsecured debt prior to, and after, the firm's IPO

	(1) Before IPO	(2) After IPO
Secured	328	265
Unsecured	44	51
Unspecified	147	158
Total number of loans	519	474

Entries in panel A show the number of firms without an S&P senior debt rating at the time the loan was taken, for both pre- and post-IPO loans (column 1), as well as the number of firms with a debt rating before and after the IPO (column 2). For loans with a debt rating, the rating is classified as Above BBB (column 3), Vulnerable (column 4), and Below Vulnerable (column 5). Entries in panel B show S&P senior debt ratings at the onset of the loan, classified for whether the loan is pre- or post-IPO. Entries in panel C show the number of loans in the final sample that are secured and unsecured (for several loans it is unspecified whether they are secured or not, and no inference can be made, since the source of ambiguity originates in the loan agreement itself). Loans are classified by pre- and post-IPO. Information on whether the loan is secured or unsecured is from the DukaBank database.

Table 3
(Continued)

Panel C: More relationship variables

	Lead lender (1)	Lender (2)
Relationship variables		
Number of prior loans by lead	-52.70**	-38.40
Total number of loans to date	(25.18)	(24.04)
More relationship variables		
Kept immediate prior lead lender	2.74 (11.79)	-22.14 (16.24)
Kept any immediate prior lender		
Loan characteristics		
First loan	-47.66*	-61.65**
Number of lending banks	(25.66)	(26.63)
Loan purpose and type	-2.59*	-2.57*
Debt repayment	(1.06)	(1.05)
LBOMBO	9.78	7.70
Takeover	(14.17)	(14.10)
364-day facility	67.51	53.37
Revolver	(52.17)	(51.89)
Term loan	68.69*	68.19**
Firm fixed effects	(26.82)	(26.71)
Loan year fixed effects	-14.52	-15.49
R ² within	(19.06)	(19.00)
F-statistic	-31.14**	-32.20**
Number of firms	(11.55)	(11.55)
Number of loans	-37.88**	-36.00*
Average number of loans	Yes	Yes
Max. number of loans per firm	Yes	Yes
Min. number of loans per firm	Yes	Yes

Only post-IPO loans, and only firms with a minimum of two such loans, are included in the final sample. Results correspond to fixed effects estimates of:

$$\text{Spread}_{i,t} = \beta_0 + \beta_1 \text{Intensity}_{i,t} + \beta_2 \text{LoanAge}_{i,t} + \beta_3 \text{Switched}_{i,t} + \beta_4 \text{LoanCharacteristics}_{i,t} + \beta_5 \text{FirmCharacteristics}_{i,t} + \beta_6 \text{LoanYear}_{i,t} + \beta_7 \text{LoanType}_{i,t} + \beta_8 \text{LoanPurpose}_{i,t} + \beta_9 \text{LoanMaturity}_{i,t} + \beta_{10} \text{LoanLeverage}_{i,t} + \beta_{11} \text{LoanRating}_{i,t} + \beta_{12} \text{LoanType}_{i,t} + \beta_{13} \text{LoanPurpose}_{i,t} + \beta_{14} \text{LoanMaturity}_{i,t} + \beta_{15} \text{LoanLeverage}_{i,t} + \beta_{16} \text{LoanRating}_{i,t} + \beta_{17} \text{LoanType}_{i,t} + \beta_{18} \text{LoanPurpose}_{i,t} + \beta_{19} \text{LoanMaturity}_{i,t} + \beta_{20} \text{LoanLeverage}_{i,t} + \beta_{21} \text{LoanRating}_{i,t} + \beta_{22} \text{LoanType}_{i,t} + \beta_{23} \text{LoanPurpose}_{i,t} + \beta_{24} \text{LoanMaturity}_{i,t} + \beta_{25} \text{LoanLeverage}_{i,t} + \beta_{26} \text{LoanRating}_{i,t} + \beta_{27} \text{LoanType}_{i,t} + \beta_{28} \text{LoanPurpose}_{i,t} + \beta_{29} \text{LoanMaturity}_{i,t} + \beta_{30} \text{LoanLeverage}_{i,t} + \beta_{31} \text{LoanRating}_{i,t} + \beta_{32} \text{LoanType}_{i,t} + \beta_{33} \text{LoanPurpose}_{i,t} + \beta_{34} \text{LoanMaturity}_{i,t} + \beta_{35} \text{LoanLeverage}_{i,t} + \beta_{36} \text{LoanRating}_{i,t} + \beta_{37} \text{LoanType}_{i,t} + \beta_{38} \text{LoanPurpose}_{i,t} + \beta_{39} \text{LoanMaturity}_{i,t} + \beta_{40} \text{LoanLeverage}_{i,t} + \beta_{41} \text{LoanRating}_{i,t} + \beta_{42} \text{LoanType}_{i,t} + \beta_{43} \text{LoanPurpose}_{i,t} + \beta_{44} \text{LoanMaturity}_{i,t} + \beta_{45} \text{LoanLeverage}_{i,t} + \beta_{46} \text{LoanRating}_{i,t} + \beta_{47} \text{LoanType}_{i,t} + \beta_{48} \text{LoanPurpose}_{i,t} + \beta_{49} \text{LoanMaturity}_{i,t} + \beta_{50} \text{LoanLeverage}_{i,t} + \beta_{51} \text{LoanRating}_{i,t} + \beta_{52} \text{LoanType}_{i,t} + \beta_{53} \text{LoanPurpose}_{i,t} + \beta_{54} \text{LoanMaturity}_{i,t} + \beta_{55} \text{LoanLeverage}_{i,t} + \beta_{56} \text{LoanRating}_{i,t} + \beta_{57} \text{LoanType}_{i,t} + \beta_{58} \text{LoanPurpose}_{i,t} + \beta_{59} \text{LoanMaturity}_{i,t} + \beta_{60} \text{LoanLeverage}_{i,t} + \beta_{61} \text{LoanRating}_{i,t} + \beta_{62} \text{LoanType}_{i,t} + \beta_{63} \text{LoanPurpose}_{i,t} + \beta_{64} \text{LoanMaturity}_{i,t} + \beta_{65} \text{LoanLeverage}_{i,t} + \beta_{66} \text{LoanRating}_{i,t} + \beta_{67} \text{LoanType}_{i,t} + \beta_{68} \text{LoanPurpose}_{i,t} + \beta_{69} \text{LoanMaturity}_{i,t} + \beta_{70} \text{LoanLeverage}_{i,t} + \beta_{71} \text{LoanRating}_{i,t} + \beta_{72} \text{LoanType}_{i,t} + \beta_{73} \text{LoanPurpose}_{i,t} + \beta_{74} \text{LoanMaturity}_{i,t} + \beta_{75} \text{LoanLeverage}_{i,t} + \beta_{76} \text{LoanRating}_{i,t} + \beta_{77} \text{LoanType}_{i,t} + \beta_{78} \text{LoanPurpose}_{i,t} + \beta_{79} \text{LoanMaturity}_{i,t} + \beta_{80} \text{LoanLeverage}_{i,t} + \beta_{81} \text{LoanRating}_{i,t} + \beta_{82} \text{LoanType}_{i,t} + \beta_{83} \text{LoanPurpose}_{i,t} + \beta_{84} \text{LoanMaturity}_{i,t} + \beta_{85} \text{LoanLeverage}_{i,t} + \beta_{86} \text{LoanRating}_{i,t} + \beta_{87} \text{LoanType}_{i,t} + \beta_{88} \text{LoanPurpose}_{i,t} + \beta_{89} \text{LoanMaturity}_{i,t} + \beta_{90} \text{LoanLeverage}_{i,t} + \beta_{91} \text{LoanRating}_{i,t} + \beta_{92} \text{LoanType}_{i,t} + \beta_{93} \text{LoanPurpose}_{i,t} + \beta_{94} \text{LoanMaturity}_{i,t} + \beta_{95} \text{LoanLeverage}_{i,t} + \beta_{96} \text{LoanRating}_{i,t} + \beta_{97} \text{LoanType}_{i,t} + \beta_{98} \text{LoanPurpose}_{i,t} + \beta_{99} \text{LoanMaturity}_{i,t} + \beta_{100} \text{LoanLeverage}_{i,t} + \beta_{101} \text{LoanRating}_{i,t} + \beta_{102} \text{LoanType}_{i,t} + \beta_{103} \text{LoanPurpose}_{i,t} + \beta_{104} \text{LoanMaturity}_{i,t} + \beta_{105} \text{LoanLeverage}_{i,t} + \beta_{106} \text{LoanRating}_{i,t} + \beta_{107} \text{LoanType}_{i,t} + \beta_{108} \text{LoanPurpose}_{i,t} + \beta_{109} \text{LoanMaturity}_{i,t} + \beta_{110} \text{LoanLeverage}_{i,t} + \beta_{111} \text{LoanRating}_{i,t} + \beta_{112} \text{LoanType}_{i,t} + \beta_{113} \text{LoanPurpose}_{i,t} + \beta_{114} \text{LoanMaturity}_{i,t} + \beta_{115} \text{LoanLeverage}_{i,t} + \beta_{116} \text{LoanRating}_{i,t} + \beta_{117} \text{LoanType}_{i,t} + \beta_{118} \text{LoanPurpose}_{i,t} + \beta_{119} \text{LoanMaturity}_{i,t} + \beta_{120} \text{LoanLeverage}_{i,t} + \beta_{121} \text{LoanRating}_{i,t} + \beta_{122} \text{LoanType}_{i,t} + \beta_{123} \text{LoanPurpose}_{i,t} + \beta_{124} \text{LoanMaturity}_{i,t} + \beta_{125} \text{LoanLeverage}_{i,t} + \beta_{126} \text{LoanRating}_{i,t} + \beta_{127} \text{LoanType}_{i,t} + \beta_{128} \text{LoanPurpose}_{i,t} + \beta_{129} \text{LoanMaturity}_{i,t} + \beta_{130} \text{LoanLeverage}_{i,t} + \beta_{131} \text{LoanRating}_{i,t} + \beta_{132} \text{LoanType}_{i,t} + \beta_{133} \text{LoanPurpose}_{i,t} + \beta_{134} \text{LoanMaturity}_{i,t} + \beta_{135} \text{LoanLeverage}_{i,t} + \beta_{136} \text{LoanRating}_{i,t} + \beta_{137} \text{LoanType}_{i,t} + \beta_{138} \text{LoanPurpose}_{i,t} + \beta_{139} \text{LoanMaturity}_{i,t} + \beta_{140} \text{LoanLeverage}_{i,t} + \beta_{141} \text{LoanRating}_{i,t} + \beta_{142} \text{LoanType}_{i,t} + \beta_{143} \text{LoanPurpose}_{i,t} + \beta_{144} \text{LoanMaturity}_{i,t} + \beta_{145} \text{LoanLeverage}_{i,t} + \beta_{146} \text{LoanRating}_{i,t} + \beta_{147} \text{LoanType}_{i,t} + \beta_{148} \text{LoanPurpose}_{i,t} + \beta_{149} \text{LoanMaturity}_{i,t} + \beta_{150} \text{LoanLeverage}_{i,t} + \beta_{151} \text{LoanRating}_{i,t} + \beta_{152} \text{LoanType}_{i,t} + \beta_{153} \text{LoanPurpose}_{i,t} + \beta_{154} \text{LoanMaturity}_{i,t} + \beta_{155} \text{LoanLeverage}_{i,t} + \beta_{156} \text{LoanRating}_{i,t} 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\text{LoanMaturity}_{i,t} + \beta_{260} \text{LoanLeverage}_{i,t} + \beta_{261} \text{LoanRating}_{i,t} + \beta_{262} \text{LoanType}_{i,t} + \beta_{263} \text{LoanPurpose}_{i,t} + \beta_{264} \text{LoanMaturity}_{i,t} + \beta_{265} \text{LoanLeverage}_{i,t} + \beta_{266} \text{LoanRating}_{i,t} + \beta_{267} \text{LoanType}_{i,t} + \beta_{268} \text{LoanPurpose}_{i,t} + \beta_{269} \text{LoanMaturity}_{i,t} + \beta_{270} \text{LoanLeverage}_{i,t} + \beta_{271} \text{LoanRating}_{i,t} + \beta_{272} \text{LoanType}_{i,t} + \beta_{273} \text{LoanPurpose}_{i,t} + \beta_{274} \text{LoanMaturity}_{i,t} + \beta_{275} \text{LoanLeverage}_{i,t} + \beta_{276} \text{LoanRating}_{i,t} + \beta_{277} \text{LoanType}_{i,t} + \beta_{278} \text{LoanPurpose}_{i,t} + \beta_{279} \text{LoanMaturity}_{i,t} + \beta_{280} \text{LoanLeverage}_{i,t} + \beta_{281} \text{LoanRating}_{i,t} + \beta_{282} \text{LoanType}_{i,t} + \beta_{283} \text{LoanPurpose}_{i,t} + \beta_{284} \text{LoanMaturity}_{i,t} + \beta_{285} \text{LoanLeverage}_{i,t} + \beta_{286} \text{LoanRating}_{i,t} + \beta_{287} \text{LoanType}_{i,t} + \beta_{288} \text{LoanPurpose}_{i,t} + \beta_{289} \text{LoanMaturity}_{i,t} + \beta_{290} \text{LoanLeverage}_{i,t} + \beta_{291} \text{LoanRating}_{i,t} + \beta_{292} \text{LoanType}_{i,t} + \beta_{293} \text{LoanPurpose}_{i,t} + \beta_{294} \text{LoanMaturity}_{i,t} + \beta_{295} \text{LoanLeverage}_{i,t} + \beta_{296} \text{LoanRating}_{i,t} + \beta_{297} \text{LoanType}_{i,t} + \beta_{298} \text{LoanPurpose}_{i,t} + \beta_{299} \text{LoanMaturity}_{i,t} + \beta_{300} \text{LoanLeverage}_{i,t} + \beta_{301} \text{LoanRating}_{i,t} + \beta_{302} \text{LoanType}_{i,t} + \beta_{303} \text{LoanPurpose}_{i,t} + \beta_{304} \text{LoanMaturity}_{i,t} + \beta_{305} \text{LoanLeverage}_{i,t} + \beta_{306} \text{LoanRating}_{i,t} + \beta_{307} \text{LoanType}_{i,t} + \beta_{308} \text{LoanPurpose}_{i,t} + \beta_{309} \text{LoanMaturity}_{i,t} + \beta_{310} \text{LoanLeverage}_{i,t} + \beta_{311} \text{LoanRating}_{i,t} + \beta_{312} \text{LoanType}_{i,t} + \beta_{313} \text{LoanPurpose}_{i,t} + \beta_{314} \text{LoanMaturity}_{i,t} + \beta_{315} \text{LoanLeverage}_{i,t} + \beta_{316} \text{LoanRating}_{i,t} + \beta_{317} \text{LoanType}_{i,t} + \beta_{318} \text{LoanPurpose}_{i,t} + \beta_{319} \text{LoanMaturity}_{i,t} + \beta_{320} \text{LoanLeverage}_{i,t} + \beta_{321} \text{LoanRating}_{i,t} + \beta_{322} \text{LoanType}_{i,t} + \beta_{323} \text{LoanPurpose}_{i,t} + \beta_{324} \text{LoanMaturity}_{i,t} + \beta_{325} \text{LoanLeverage}_{i,t} + \beta_{326} \text{LoanRating}_{i,t} + \beta_{327} \text{LoanType}_{i,t} + \beta_{328} \text{LoanPurpose}_{i,t} + \beta_{329} \text{LoanMaturity}_{i,t} + \beta_{330} \text{LoanLeverage}_{i,t} + \beta_{331} \text{LoanRating}_{i,t} + \beta_{332} \text{LoanType}_{i,t} + \beta_{333} \text{LoanPurpose}_{i,t} + \beta_{334} \text{LoanMaturity}_{i,t} + \beta_{335} \text{LoanLeverage}_{i,t} + \beta_{336} \text{LoanRating}_{i,t} + \beta_{337} \text{LoanType}_{i,t} + \beta_{338} \text{LoanPurpose}_{i,t} + \beta_{339} \text{LoanMaturity}_{i,t} + \beta_{340} \text{LoanLeverage}_{i,t} + \beta_{341} \text{LoanRating}_{i,t} + \beta_{342} \text{LoanType}_{i,t} + \beta_{343} \text{LoanPurpose}_{i,t} + \beta_{344} \text{LoanMaturity}_{i,t} + \beta_{345} \text{LoanLeverage}_{i,t} + \beta_{346} \text{LoanRating}_{i,t} + \beta_{347} \text{LoanType}_{i,t} + \beta_{348} \text{LoanPurpose}_{i,t} + \beta_{349} \text{LoanMaturity}_{i,t} + \beta_{350} \text{LoanLeverage}_{i,t} + \beta_{351} \text{LoanRating}_{i,t} + \beta_{352} \text{LoanType}_{i,t} + \beta_{353} \text{LoanPurpose}_{i,t} + \beta_{354} \text{LoanMaturity}_{i,t} + \beta_{355} \text{LoanLeverage}_{i,t} + \beta_{356} \text{LoanRating}_{i,t} + \beta_{357} \text{LoanType}_{i,t} + \beta_{358} \text{LoanPurpose}_{i,t} + \beta_{359} \text{LoanMaturity}_{i,t} + \beta_{360} \text{LoanLeverage}_{i,t} + \beta_{361} \text{LoanRating}_{i,t} + \beta_{362} \text{LoanType}_{i,t} + \beta_{363} \text{LoanPurpose}_{i,t} + \beta_{364} \text{LoanMaturity}_{i,t} + \beta_{365} \text{LoanLeverage}_{i,t} + \beta_{366} \text{LoanRating}_{i,t} + \beta_{367} \text{LoanType}_{i,t} + \beta_{368} \text{LoanPurpose}_{i,t} + \beta_{369} \text{LoanMaturity}_{i,t} + \beta_{370} \text{LoanLeverage}_{i,t} + \beta_{371} \text{LoanRating}_{i,t} + \beta_{372} \text{LoanType}_{i,t} + \beta_{373} \text{LoanPurpose}_{i,t} + \beta_{374} \text{LoanMaturity}_{i,t} + \beta_{375} \text{LoanLeverage}_{i,t} + \beta_{376} \text{LoanRating}_{i,t} + \beta_{377} \text{LoanType}_{i,t} + \beta_{378} \text{LoanPurpose}_{i,t} + \beta_{379} \text{LoanMaturity}_{i,t} + \beta_{380} \text{LoanLeverage}_{i,t} + \beta_{381} \text{LoanRating}_{i,t} + \beta_{382} \text{LoanType}_{i,t} + \beta_{383} \text{LoanPurpose}_{i,t} + \beta_{384} \text{LoanMaturity}_{i,t} + \beta_{385} \text{LoanLeverage}_{i,t} + \beta_{386} \text{LoanRating}_{i,t} + \beta_{387} \text{LoanType}_{i,t} + \beta_{388} \text{LoanPurpose}_{i,t} + \beta_{389} \text{LoanMaturity}_{i,t} + \beta_{390} \text{LoanLeverage}_{i,t} + \beta_{391} \text{LoanRating}_{i,t} + \beta_{392} \text{LoanType}_{i,t} + \beta_{393} \text{LoanPurpose}_{i,t} + \beta_{394} \text{LoanMaturity}_{i,t} + \beta_{395} \text{LoanLeverage}_{i,t} + \beta_{396} \text{LoanRating}_{i,t} + \beta_{397} \text{LoanType}_{i,t} + \beta_{398} \text{LoanPurpose}_{i,t} + \beta_{399} \text{LoanMaturity}_{i,t} + \beta_{40$$

- **Takeaway.** The change in pricing around the IPO is not just a change in borrower risk; it is a change in how lenders price a relationship once information becomes public.

5.5 Table 6: Alternative relationship-intensity measure

Table 6
Robustness analysis: Alternative measure of relationship intensity

	Baseline (1)	Rating (2)	Maturity (3)	Loan purpose (4)	Firm age (5)
Relationship intensity					
Intensity.Amount * After.IPO	-31.35** (15.03)	-28.48* (14.87)	-26.53* (14.83)	-23.79* (14.47)	-26.85* (14.86)
Intensity.Amount * Before.IPO	27.91* (15.67)	32.07** (15.53)	32.52** (15.46)	30.41** (15.08)	32.04** (15.51)
Loan characteristics					
Loan number	13.12*** (2.68)	12.74*** (2.65)	12.50*** (2.64)	13.55*** (2.59)	12.64*** (2.67)
Ln (Loan amount)	-5.04 (4.24)	-4.79 (4.19)	-7.54* (4.33)	-9.09** (4.23)	-7.32* (4.37)
SP Rating: Vulnerable		122.03*** (33.45)	114.74*** (33.45)	115.40*** (32.64)	113.17*** (33.67)
Loan maturity			0.36** (0.15)	0.25* (0.15)	0.36** (0.15)
Loan purpose: LBO/MBO				103.71*** (19.56)	
Loan purpose: Takeover				21.12 (15.17)	
Firm characteristics					
Firm age at time of loan					-1.11 (2.57)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Loan year fixed effects	Yes	Yes	Yes	Yes	Yes
R ² within	0.12	0.14	0.15	0.20	0.15
F-statistic	7.06	7.78	7.66	8.97	7.07
Number of loans	878	878	874	874	874
Number of firms	358	358	354	354	354
Average number of loans per firm	2.45	2.45	2.47	2.47	2.47

The reported results correspond to fixed effects estimates of regression equations of the type:

$$\begin{aligned}
 Spread_{i,l} = & \beta_0 + \beta_{Intensity.Amount.After} (Intensity.Amount_{i,l} * After.IPO_{i,l}) \\
 & + \beta_{Intensity.Amount.Before} (Intensity.Amount_{i,l} * Before.IPO_{i,l}) + \beta_{Switched} Switched_{i,l} \\
 & + \beta_{Loan.Characteristics} Loan.Characteristics_{i,l} + \beta_{Firm.Characteristics} Firm.Characteristics_{i,l} \\
 & + \beta_{Loan.Yrs} Loan.years.FE_{i,l} + \varepsilon_i + \mu_{i,l},
 \end{aligned}$$

where i indexes for firm and l for the loan number.

$$Intensity.Amount_{i,l} = \left(\frac{Total\ amount\ borrowed\ from\ current\ lead\ lender\ to\ date}{Total\ amount\ borrowed\ to\ date} \right)_{i,l}$$

$Spread_{i,l}$ is the all-in-spread drawn for firm i 's loan l . $Switched.Lenders_{i,l}$ is a categorical variable equal to 1 when the firm drops all previous lenders and borrows from a new lender. $Loan.Yrs.FE_{i,l}$ is a vector of firm i 's loan l 's characteristics, such as the number of lenders in the lending syndicate, the time to maturity (in months), and the type and purpose of the loan. $Firm.Characteristics_{i,l}$ is a vector of firm i 's characteristics at the time loan l was taken, such as the debt rating firm i received when taking loan l . Debt ratings are S&P senior debt ratings at the onset of the loan. When a loan has several facilities, the spread and fees are the weighted average of the spread of each facility, where the weights are the amount of the loan in that facility relative to the total loan amount. Also in the case of several facilities, the time to maturity is the maximum between the maturities of each facility; and the amount of the loan is the sum of the amounts in each facility. Standard errors are reported in parentheses, below the coefficient. I use ***, **, and * to denote significance at the 1%, 5%, and 10% levels, respectively.

- **Read:** This table replaces the count-based intensity measure with a dollar-based dependence measure.
- Pre-IPO, the linear coefficient is positive, ranging from about **27.91** to **32.52**.
- Post-IPO, the coefficient is negative, ranging from about **-23.79** to **-31.35**.
- **Takeaway.** The main result is not an artifact of counting loans rather than measuring exposure by loan size.

5.6 Table 7 and Table 8: Risk sharing versus information rents

- **Read:** Table 7 shows that larger syndicates and smaller lead-lender shares are correlated with lower spreads. Table 8 asks whether these variables explain away the relationship-intensity patterns.
- In simple regressions, more lenders are associated with slightly lower spreads, and a higher lead-lender share is associated with higher spreads.

Table 7
The lead lender's risk sharing ability and the firm's cost of borrowing pre- and post-IPO

	Number of lenders		Percent lead lender lent		HHI	
	(1)	(2)	(3)	(4)	(5)	(6)
Risk sharing measures						
Number of lenders	-1.58** (0.63)	-1.37* (0.77)				
Number of lenders * After IPO		-0.47 (0.98)				
Percent lead lent			0.59** (0.29)	0.61** (0.30)		
Percent lead lent * After IPO				-0.05 (0.20)		
HHI					19.86 (20.56)	21.26 (24.21)
HHI * After IPO					-2.88 (26.18)	-51.08* (26.46)
After IPO	-40.81*** (10.58)	-37.67*** (12.43)	-45.45*** (11.95)	-42.48*** (16.01)	-53.14*** (18.56)	-51.08* (26.46)
Firm and loan characteristics						
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Loan year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of loans	660	660	553	553	308	308
Number of firms	159	159	134	134	108	108
R ² within	0.07	0.07	0.06	0.06	0.10	0.10
F-statistic	4.89	4.36	3.51	3.13	2.67	2.36
Average nu of loans per firm	4.15	4.15	4.13	4.13	2.85	2.85

The reported results correspond to fixed effects estimates of regression equations of the type:

$$\begin{aligned}
 Spread_{i,l} = & \beta_0 + \beta_{Risk.Sharing} Risk.Sharing_{i,l} + \beta_{Risk.Sharing.After} Risk.Sharing.After_{i,l} \\
 & + \beta_{After.IPO} After.IPO_{i,l} + \varepsilon_i + \mu_{i,l},
 \end{aligned}$$

where i indexes for firm and l for the loan number. $Spread_{i,l}$ is the all-in-spread drawn for firm i 's loan l ; and where $Risk.Sharing_{i,l}$ is one of the three risk-sharing measures: In columns 1 and 2 the risk-sharing variable is the number of lenders in the lending syndicate: $Nu.Lenders_{i,l}$ equals the number of banks in the lending syndicate of loan l for firm i . In Columns 3 and 4 the risk-sharing variable is $Pct.Lead.Lent_{i,l}$, which for each firm i 's loan l is the percentage of the loan amount contributed by the lead lender. In Columns 5 and 6 the risk-sharing measure is the Herfindahl index, equal to the sum of the squared percentage that each lender contributed to the loan. $Risk.Sharing.After_{i,l} = Risk.Sharing_{i,l} * After.IPO_{i,l}$. All regressions include firm fixed effects and relationship year fixed effects. Data for the risk-sharing measures have been collected from the Dealscan database. Standard errors are reported in parentheses, below the coefficient. I use ***, **, and * to denote significance at the 1%, 5%, and 10% levels, respectively.

Table 8
Alternative hypothesis: The lead lender's risk sharing ability, relationship intensity, and the firm's cost of borrowing pre- and post-IPO

	No lenders			Pct lead lent			
	(1)	* After IPO	(2)	(3)	* After IPO	Lead = 100	No lead = 100
					(4)	(5)	(6)
Relationship intensity							
Number of prior loans by lead	-73.58***	-72.27**	-99.47***	-100.02***	-99.45***	-103.05**	
• After IPO	(28.17)	(28.87)	(34.16)	(34.55)	(34.21)	(40.86)	
Total number of loans to date							
Number of prior loans by lead	246.52**	246.06**	-317.17**	-316.07**	-317.55**	-352.22**	
• Before IPO	(105.11)	(105.23)	(130.27)	(130.80)	(130.67)	(157.75)	
Total number of loans to date	207.02*	206.32*	255.57*	254.26	255.84*	324.13*	
(Number of prior loans by lead							
• Before-IPO) ²	(118.68)	(118.84)	(154.38)	(155.02)	(154.69)	(174.72)	
Switched lenders	-49.99***	-49.63***	-65.03***	-65.01***	-65.05***	-66.02**	
	(17.35)	(17.63)	(21.00)	(21.03)	(21.03)	(26.39)	
Loan characteristics							
First loan	-62.52**	-61.96**	-78.09**	-78.23**	-78.21**	-80.22**	
	(25.78)	(25.94)	(30.45)	(30.52)	(30.50)	(39.10)	
After IPO	-61.67***	-60.58***	-64.41***	-62.47***	-64.57***	-61.92*	
	(20.18)	(20.84)	(23.81)	(29.25)	(24.04)	(32.23)	
Loan characteristics and risk-sharing measures							
Number of lenders	-1.38*	-1.28					
	(0.64)	(0.78)					
Number of lenders * After IPO		-0.72					
		(0.28)					
Percent lead lent		0.47	0.48	0.49	0.49	0.64	
		(0.33)	(0.35)	(0.50)	(0.50)	(0.52)	
Percent lead lent * After IPO		-0.03	-0.03	-0.03	-0.03	-0.03	
		(0.25)	(0.25)	(0.25)	(0.25)	(0.25)	
Percent lead lent = 100					-1.72	(33.97)	

(continued overleaf)

Table 8
(Continued.)

	No lenders			Pct lead lent		
	(1)	* After IPO (2)	(3)	* After IPO (4)	Lead = 100 (5)	No lead = 100 (6)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Loan year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of loans	660	660	491	491	491	297
Number of firms	159	159	126	126	126	65
R ² within	0.09	0.09	0.08	0.08	0.08	0.11
F-statistic	3.75	3.46	2.46	2.47	2.47	2.28
Average number of loans per firm	4.15	4.15	3.90	3.90	3.90	4.57
Min. number of loans per firm	2	2	2	2	2	2
Max. number of loans per firm	14	14	14	14	14	12

The reported results correspond to fixed effects estimates of regression equations of the type:

$$\text{Spread}_{i,t} = \beta_0 + \beta_{\text{Intensity},i,t}(\text{Intensity}_{i,t} * \text{After-IPO}_{i,t}) + \beta_{\text{Before-IPO},i,t}(\text{Before-IPO}_{i,t}) + \beta_{\text{After-IPO},i,t}(\text{After-IPO}_{i,t}) + \beta_{\text{Switched},i,t}(\text{Switched}_{i,t}) + \beta_{\text{Loan-Characteristics},i,t}(\text{Loan-Characteristics}_{i,t}) + \beta_{\text{Firm-Characteristics},i,t}(\text{Firm-Characteristics}_{i,t}) + \beta_{\text{Loan-years-FE}_{i,t}} + \epsilon_i + \eta_{i,t}$$

where i indexes for firm and t for the loan number.

$$\text{Intensity}_{i,t} = \frac{\text{Number of prior loans by lead}}{\text{Total number of loans to date}}$$

$\text{Spread}_{i,t}$ is the all-in-spread drawn for firm i 's loan t . All regressions include firm fixed effects and relationship year fixed effects. In columns 1 and 2 the risk-sharing variable is $\text{No-Leader}_{i,t}$ equal to the number of banks in the lending syndicate of firm i 's loan t . In columns 3 through 6, the risk-sharing variable is $\text{Pct-Lead-Lent}_{i,t}$, which is the percentage of firm i 's loan t amount contributed by the lead lender. Standard errors are reported in parentheses, below the coefficient. I use ***, **, and * to denote significance at the 1%, 5%, and 10% levels, respectively.

- But once the relationship-intensity controls are introduced, the risk-sharing variables lose explanatory power while the main before/after IPO pattern remains.
- **Takeaway.** Risk sharing can explain part of the raw post-IPO spread decline, but it does *not* replace the information-rent interpretation.

5.7 Table 9 and Table 10: Concurrent lending and underwriting

Table 9
Alternative hypothesis: Concurrent lending and underwriting

	Lead lender is also the	
	IPO book runner (1)	IPO mgr or co-mgr (2)
Lead lender is IPO book runner	12.57 (17.05)	
Lead lender is IPO book runner * After IPO	-9.52 (20.95)	
Lead lender is IPO Mgr		15.34 (12.69)
Lead lender is IPO mgr * After IPO		-2.16 (14.33)
After IPO	-40.45*** (11.06)	-40.84*** (11.06)
Firm fixed effects	Yes	Yes
Loan year fixed effects	Yes	Yes
R ² within	0.06	0.07
F-statistic	3.73	3.89
Average number of loans per firm	4.12	4.12
Number of loans	659	659
Number of firms	160	160

The reported results correspond to fixed effects estimates of regression equations of the type:

$$\text{Spread}_{i,t} = \beta_0 + \beta_{\text{Lender-IPO-Underwriter},i,t}(\text{Lender-IPO-Underwriter}_{i,t}) + \beta_{\text{Before-IPO},i,t}(\text{Before-IPO}_{i,t}) + \beta_{\text{After-IPO},i,t}(\text{After-IPO}_{i,t}) + \epsilon_i + \eta_{i,t}$$

where i indexes for firm and t for loan number. $\text{Spread}_{i,t}$ is the all-in-spread drawn for firm i 's loan t . In column 1, $\text{Lender-IPO-Underwriter}_{i,t}$ is a categorical variable equal to 1 if firm i 's lead lender for loan t is also the IPO manager or co-manager and 0 otherwise. In column 2, $\text{Lender-IPO-Underwriter}_{i,t}$ is a categorical variable equal to 1 when firm i 's lead lender for loan t is one of firm i 's IPO book runners, and 0 otherwise. All regressions include firm fixed effects and relationship year fixed effects. Standard errors are reported in parentheses, below the coefficient. I use ***, **, and * to denote significance at the 1%, 5%, and 10% levels, respectively.

Table 10
Alternative hypothesis: Concurrent lending and underwriting

	Lead lender is also the			
	IPO manager or co-manager		IPO bookrunner	
	(1)	(2)	(3)	(4)
Relationship intensity				
Number of prior loans by lead	-72.92***	-73.17***	-70.63***	-70.39**
Total number of loans to date	(27.29)	(27.55)	(27.16)	(27.28)
Number of prior loans by lead	-225.66**	-252.44**	-252.14**	-222.83**
Total number of loans to date	(101.34)	(101.49)	(101.57)	(101.81)
(Number of prior loans by lead Total number of loans to date * Before-IPO)	180.16*	197.16*	219.92**	193.94**
Switched lenders	(114.18)	(114.41)	(114.78)	(115.14)
Loan characteristics	-13.53***	-13.53***	-54.08***	-54.08***
After IPO	(16.96)	(17.02)	(16.98)	(17.00)
Relation between lead lender and IPO underwriter	8.55	8.03		
Lead lender is IPO mgr	(10.09)	(12.54)		
Lead lender is IPO mgr * After IPO	1.01	(14.34)		
Lead lender is IPO book runner			4.75	5.88
Lead lender is IPO book runner * After IPO			(14.27)	(16.69)
Loan characteristics			-2.66	-2.66
After IPO			(20.51)	(20.51)
First loan	-64.86***	-65.20***	-65.41***	-65.16***
After IPO	(19.82)	(20.41)	(19.83)	(19.84)
Number of lending bank	-66.72***	-66.89***	-66.72***	-66.63***
Loan purpose: LBO,MBO	(25.07)	(25.21)	(25.06)	(25.12)
Loan purpose: Takeover * After IPO	-1.58**	-1.58**	-1.61**	-1.61**
Loan purpose: Revolver	(0.62)	(0.62)	(0.62)	(0.63)
Loan risk	79.86***	79.93***	81.03***	80.92***
Above BBB	(19.22)	(19.21)	(19.17)	(19.21)
Vulnerable	44.28*	44.24*	43.77*	43.92*
Firm fixed effects	Yes	Yes	Yes	Yes
Loan year fixed effects	Yes	Yes	Yes	Yes
R ² within	0.18	0.18	0.18	0.18
F-statistic	5.08	4.80	5.01	4.77
Average number of loans per firm	4.13	4.13	4.13	4.13
Number of loans	657	657	657	657
Number of firms	159	159	159	159

Table 10
(Continued)

	Exclude loans where the lead lender is					
	Also the IPO manager gr bookrunner			Also the IPO manager gr bookrunner		
	(1)	(2)	(3)	(4)	(5)	(6)
Relationship intensity						
Number of prior loans by lead	-61.16**	-58.82**	-61.06**	-56.83**	-54.68**	-51.27**
Total number of loans to date	(28.20)	(27.96)	(27.59)	(25.18)	(25.03)	(24.90)
Number of prior loans by lead	-211.63**	-217.59**	-228.22**	-242.61**	-224.41**	-218.44**
Total number of loans to date	(91.73)	(90.66)	(89.65)	(82.87)	(82.30)	(81.55)
(Number of prior loans by lead Total number of loans to date * Before-IPO)	180.16*	197.16*	219.92**	193.94**	187.80**	170.08*
Switched lenders	(109.42)	(108.34)	(106.91)	(96.80)	(96.00)	(95.02)
Loan characteristics	-27.66*	-27.66*	-31.46**	-32.30**	-32.30**	-33.91**
After IPO	(16.12)	(15.93)	(15.74)	(14.81)	(14.51)	(14.37)
First loan	-44.65***	-42.16***	-44.61***	-39.78**	-49.08***	-49.30***
After IPO	(14.85)	(14.69)	(14.57)	(13.14)	(13.09)	(12.99)
Number of lending banks	-38.00**	-40.94**	-45.37**	-38.80**	-35.09**	-34.40**
Loan maturity	(20.75)	(20.53)	(20.31)	(18.45)	(18.33)	(18.32)
Loan purpose: LBO,MBO	-0.35	-0.23	-0.04	-0.77	-0.84	-0.79
S & P rating: Above BBB	(0.66)	(0.66)	(0.65)	(0.53)	(0.53)	(0.52)
S & P rating: Vulnerable	0.49***	0.42***	0.36**	0.36**	0.32**	0.28**
Loan purpose: LBO,MBO	(0.14)	(0.14)	(0.14)	(0.13)	(0.13)	(0.13)
S & P rating: Above BBB	65.00***	59.71***	65.00***	59.71***	59.71***	51.16***
S & P rating: Vulnerable	(19.25)	(19.25)	(19.25)	(18.10)	(18.10)	(18.10)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Loan year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of loans	714	714	714	832	832	832
Number of firms	314	314	314	330	330	330
Average number of loans per firm	2.32	2.32	2.32	2.52	2.52	2.52
R ² within	0.10	0.13	0.16	0.10	0.12	0.14

Only firms with pre- and post-IPO loans are considered. Each firm must have at least one loan before and one loan after the IPO. Results correspond to fixed effects estimates of

$$\text{Spread}_{i,t} = \beta_0 + \beta_{\text{Intensity},i,t}(\text{Intensity}_{i,t} * \text{After-IPO}_{i,t}) + \beta_{\text{Before-IPO},i,t}(\text{Before-IPO}_{i,t}) + \beta_{\text{After-IPO},i,t}(\text{After-IPO}_{i,t}) + \beta_{\text{Switched},i,t}(\text{Switched}_{i,t}) + \beta_{\text{Loan-Characteristics},i,t}(\text{Loan-Characteristics}_{i,t}) + \beta_{\text{Firm-Characteristics},i,t}(\text{Firm-Characteristics}_{i,t}) + \beta_{\text{Loan-years-FE}_{i,t}} + \epsilon_i + \eta_{i,t}$$

where i indexes for firm and t for the loan number.

$$\text{Intensity}_{i,t} = \frac{\text{Number of prior loans by lead}}{\text{Total number of loans to date}}$$

$\text{Spread}_{i,t}$ is the all-in-spread drawn for firm i 's loan t . All regressions include firm fixed effects and relationship year fixed effects. Panel A: columns 1 and 2 control for loans where the lead lender is also the IPO manager or co-manager; columns 3 and 4 control for loans where the lead lender is the IPO book runner. Panel B: columns 1–3 exclude loans where the lead lender is also an IPO book runner or an IPO manager; and column 4 exclude those for which the lead lender is also an IPO book runner and an IPO manager. Standard errors are reported in parentheses, below the coefficient. I use ***, **, and * to denote significance at the 1%, 5%, and 10% levels, respectively.

- **Read:** These tables test whether banks give cheaper loans in exchange for IPO underwriting mandates.
- The raw data show lower spreads when the lead lender is also an IPO manager/co-manager or book runner.
- However, once the underwriting-role indicators and their interactions with AfterIPO are added, they do not explain the spread pattern that the paper attributes to information rents.

- Even excluding loans where the lead lender also underwrites the IPO, the main relationship-intensity results remain.
- **Takeaway.** The core result is not a bundling story about loans and underwriting services.

5.8 Table 11 and Table 12: Bubble firms and prior debt issuance

Table 11
Further robustness analysis: Internet stocks, tech stocks, and the IPO bubble years

	Include		Exclude	
	Tech stocks (1)	Internet stocks (2)	Internet stocks (3)	Tech stocks (4)
Relationship intensity				
Number of prior loans by lead				
Total number of loans to date	• After-IPO	−80.28*** (27.05)	−79.57*** (26.99)	−91.06* (52.14)
Number of prior loans by lead				
Total number of loans to date	• Before-IPO	−267.06*** (101.06)	−268.88*** (100.78)	−693.76*** (210.88)
(Number of prior loans by lead Total number of loans to date)²	• Before-IPO	218.62* (114.61)	219.75* (114.30)	898.75*** (280.27)
Switched lenders		−53.73*** (16.87)	−53.93*** (16.82)	−89.28*** (30.92)
Loan characteristics				
First loan		−69.42*** (24.75)	−72.50*** (24.75)	−117.79*** (43.42)
After IPO		−64.33*** (19.24)	−65.49*** (19.17)	−55.53* (33.80)
Tech and Internet stocks				
Technology stock		63.32* (37.80)	No	No
Internet stock			68.06** (29.73)	No
Firm fixed effects	Yes	Yes	Yes	Yes
Loan year fixed effects	Yes	Yes	Yes	Yes
R-squared	0.09	0.09	0.10	0.13
F-statistic	3.91	4.11	1.69	1.33
R² within	0.09	0.09	0.10	0.13
Number of firms	701	701	221	149
Number of loans	160	160	50	41
Average number of loans per firm	4.38	4.38	4.42	3.63
Max. number of loans per firm	701	701	221	149
Min. number of loans per firm	160	160	50	41

For the purpose of this estimation, only those firms with pre- and post-IPO loans are considered. Each firm must have at least one loan before and one loan after the IPO. The reported results correspond to fixed effects estimates of

$$\begin{aligned}
 \text{Spread}_{i,l} = & \beta_0 + \beta_{\text{Intensity_After}}(\text{Intensity}_{i,l} * \text{After_IPO}_{i,l}) + \beta_{\text{Intensity_Before}}(\text{Intensity}_{i,l} * \text{Before_IPO}_{i,l}) \\
 & + \beta_{\text{Intensity_Before_Spd}}(\text{Intensity}_{i,l} * \text{Before_IPO}_{i,l}) + \beta_{\text{Switched}}\text{Switched}_{i,l} \\
 & + \beta_{\text{Loan_Characteristics}}\text{Loan_Characteristics}_{i,l} \\
 & + \beta_{\text{Firm_Characteristics}}\text{Firm_Characteristics}_{i,l} + \beta_{\text{Loan_Yrs}}\text{Loan_years}_{i,l} + \varepsilon_i + \mu_{i,l},
 \end{aligned}$$

where i indexes for firm and l for the loan number.

$$\text{Intensity}_{i,l} = \frac{\text{Number of prior loans by Lead}}{\text{Total number of loans to date}}.$$

$\text{Spread}_{i,l}$ is the all-in-spread drawn for firm i 's loan l . All regressions include firm fixed effects and relationship year fixed effects. Columns 1 and 2 control for whether the lead IPO firm is a technology stock or an Internet stock, respectively; columns 3 and 4 exclude technology and Internet stocks, respectively. Standard errors are reported in parentheses, below the coefficient. I use ***, **, and * to denote significance at the 1%, 5%, and 10% levels, respectively.

- **Read:** Table 11 checks whether the internet and tech IPO wave drives the results. Table 12 checks whether prior public or private debt issuance already revealed information before the IPO.
- Tech and internet IPOs do tend to pay higher rates on average, but including or excluding them leaves the main before/after pattern intact.
- Only **18** firms in the relevant sample have any public debt issue around the IPO window, and only **3** have only pre-IPO public debt issues. For private debt, **84** firms have any issue, but only **19** have any pre-IPO private placement.
- Excluding firms with pre-IPO public or private debt issues does not materially change the results.
- **Takeaway.** For most firms, the IPO is indeed the main information-releasing event in the sample.

Table 12
Public and private debt issues

Panel A: Public debt issues						
	Total	No public debt issues	Public debt issues	Firms with public debt issues		
Number of firms	378			Only Pre-IPO	Only Post-IPO	Pre-and post-IPO
		360	18	3	14	1
Panel B: Private debt issues						
	Total	No private debt issues	Private debt issues	Firms with private debt issues		
Number of firms	378			Only Pre-IPO	Only Post-IPO	Pre-and post-IPO
		294	84	13	65	6
Panel C: Public and private debt issues						
	No private or public issues		Private and public debt issues			
			11			
	Classified by pre- and post-IPO					
Number of firms	378	367	Pre-IPO (3 firms)		Post-IPO (11 firms)	
			Public and private	Public but not private	Public and private	Public but not private
			1	0	1	10
						1
						0

From the SDC database, I obtain all public and private debt issues between January 1, 1990, and December 31, 2004. I merge this data with the sample of firms with bank loans for which all loan characteristics are available (the sample of firms used in Tables 2–4). Panel A shows that only 19 of the sample firms have issued public debt. Of these, only three firms have pre-IPO public issues and one firm has both pre- and post-IPO public debt issues. The remaining 14 firms have post-IPO public debt issues. Panel B shows that 84 of the sample firms have issued private debt. Of these, only 13 firms have pre-IPO private issues and six firms have both pre- and post-IPO private debt issues. The remaining 65 firms have both pre- and post-IPO private debt issues. Panel C reports that only 11 firms have both private and public debt issues. Of these only one has pre-IPO issues.

5.9 Figures

- **Note.** The paper's empirical evidence is overwhelmingly **table-based**. There are no central figures carrying the main identification or event-study results in the way many modern empirical papers do.
- **Implication.** To read the paper well, one should focus on the logic of Tables 1–12 and especially on how the coefficients on relationship intensity change between the pre-IPO and post-IPO regimes.

6 Short summary

- **Method.** Use the IPO as an information shock and estimate fixed-effects loan-spread regressions with relationship-intensity measures before and after the IPO.
- **Identification.** The IPO releases information to outside lenders; firm fixed effects absorb permanent borrower heterogeneity; loan-year fixed effects absorb common credit-market conditions.
- **Application.** Syndicated loans to IPO firms from SDC matched to Dealscan, with leverage and risk controls from COMPUSTAT and IPO prospectuses.
- **Main result.** Before the IPO, loan spreads are U-shaped in relationship intensity; after the IPO, they are decreasing in relationship intensity, and average borrowing costs fall materially.
- **Interpretation.** Relationship banks appear to exploit an information-based monopoly over private firms, but lose that advantage once the firm goes public.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper provides evidence that relationship banks extract **informational rents** from borrowers before those borrowers become public.
- The evidence comes from a sharp change in how loan spreads depend on relationship intensity once the IPO releases information to the market.
- This pricing pattern is not well explained by changing leverage, changing ratings, risk sharing in syndicates, or concurrent underwriting.

7.2 Economic intuition

- At low levels of relationship intensity, repeated lending lowers costs and information can spill over to rival banks, so spreads can fall.
- At high levels of dependence before the IPO, the incumbent bank's information-processing advantage dominates, outside lenders face a stronger adverse-selection problem, and the borrower becomes locked in.

- The IPO weakens that lock-in because outside lenders now have access to audited disclosures and market prices.

7.3 Takeaway

This paper's central lesson is that the value of relationship lending is not purely benign. The same relationship that lowers monitoring and screening costs can also create an information monopoly for the incumbent lender. Public disclosure through the IPO weakens that monopoly, lowers spreads, and makes the pricing of bank loans more competitive.